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Genetic structure of a native cyprinid in a reservoir-altered stream network

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SUMMARY

1. Reservoirs modify riverine ecosystems worldwide, and often with deleterious impacts on native biota. The immediate effects of reservoirs on native fish species below dams and in impounded reaches have received considerable attention, but it is unclear how reservoirs may affect fish species at larger spatial and temporal scales. Documented declines of stream fish populations in direct tributaries of reservoirs suggest reservoir pools may reduce gene flow among historically connected populations.
2. Because of increased predator densities in reservoirs and the extent of habitat alteration in impounded reaches, I predicted reservoir habitats would reduce gene flow among small-bodied fish populations separated by reservoir habitat. I used microsatellite markers to assess the spatial genetic structure of populations of the red shiner (*Cyprinella lutrensis*), in a reservoir-fragmented stream network (Lake Texoma, U.S.A.). I also tested the prediction that populations in two direct tributaries that have experienced population declines would have low genetic diversity. Individuals were collected from six sites upstream of the reservoir, three sites in the reservoir and three sites in direct tributaries of the reservoir during 2008 and 2009.
3. Results indicate that most populations were isolated by distance with little divergence among populations. In one direct tributary population, however, there was substantial genetic divergence, and genetic diversity was significantly lower than in other populations. Gene flow also seemed to be lower in reservoir habitats than in intact stream habitats, suggesting reservoir habitats may be reducing gene flow among the reservoir-separated populations. These results indicate that reservoirs may reduce gene flow among reservoir-fragmented stream fish populations, altering the evolutionary trajectories of fragmented populations.

Keywords: barriers, *Cyprinella*, dams, population fragmentation, prairie streams

Introduction

Predicting consequences of habitat fragmentation on ecological systems is a major challenge for conservation biologists (Tilman *et al*., 1994). Riverine impoundments fragment lotic ecosystems worldwide (Nilsson & Berggren, 2000), frequently with deleterious impacts on aquatic systems at multiple spatial and temporal scales (Benke, 1990; Ward, 1998; Pringle, Freeman & Freeman, 2000; Poff *et al*., 2007). Numerous immediate and adverse effects of impoundments are relatively well documented, but it is not clear how reservoirs affect biota over larger temporal and spatial scales (Fullerton *et al*., 2010).

The degree to which impoundments alter natural stream habitats is typically implicated as the driving factor behind changes to native fish communities in reservoir-altered systems. More often than not, conversion of lotic riverine habitats to lentic reservoirs results in fish community shifts through extirpation of riverine species (Holden & Stalnaker, 1975; Martinez *et al*., 1994), increased abundance of native habitat generalists (Ruhr, 1957; Gido & Matthews, 2000; Edds, Matthews & Gelwick, 2002; Herbert & Gelwick, 2003) and increased densities of native and introduced piscivores (Martinez *et al*.,1994; Edds *et al*., 2002). Fish communities downstream of impoundments are similarly altered by extirpations or

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introductions caused by changes in temperature, flow regime, sediment loads and turbidity levels (Vanicek, Kramer & Franklin, 1970; Holden & Stalnaker, 1975; Edwards, 1978). Although the effects of reservoirs on localised fish communities upstream and downstream of impoundments are well studied, the potential for inundated stream reaches to act as barriers to migration of stream fish species has received little attention.

Fish species inhabiting small streams in undisturbed riverine systems can exhibit natural source-sink population dynamics, with coalescing streams and mainstems serving as migration corridors among populations (Fagan *et al.*, 2002). Conversely, researchers who have documented declines or extirpation of small-bodied fish species in streams that flow directly into reservoirs (i.e. direct tributaries) posit that a lack of migration through reservoir habitats (and hence reduced rescue effects) contributes to population declines (Winston, Taylor & Pigg, 1991; Lienesch, Lutterschmidt & Schaefer, 2000; Falke & Gido, 2006; Matthews & Marsh-Mathews, 2007). Reservoir habitats could functionally reduce gene flow among once-connected populations, subjecting reservoir-fragmented populations to deleterious effects associated with genetic isolation and small population sizes (i.e. inbreeding depression, genetic drift; Vrijenhoek, 1998). Moreover, even if extirpated populations in direct tributaries can be re-colonised, subsequent reestablished populations could suffer from similar deleterious effects (e.g. founder effects; Mayr, 1942; Lande, 1988).

The extent of reservoir-based population fragmentation is likely to be modulated by how disparate reservoir habitats are compared to natural streams and by the specific ecologies of stream fish species. Stream fish species with strict habitat preferences (i.e. habitat specialists) may be most susceptible to reservoir-based habitat fragmentation because the extant reservoir pools alter habitats (Schlosser *et al.*, 2000; Herbert & Gelwick, 2003; Skalksi *et al.*, 2008). In addition, the increased density of piscivorous fish species in reservoirs may lower the suitability of reservoir habitats to act as migration corridors by increasing predation pressure on small-bodied stream fish species (Schlosser *et al.*, 2000). Indeed, reservoir-based population fragmentation of a stream habitat specialist [creek chub, *Semotilis atromaculatus* (Mitchell)] revealed population isolation and reduced genetic diversity in reservoir-fragmented populations (Skalksi *et al.*, 2008). Reservoir-isolating effects may not be limited to stream habitat specialists. Many small-bodied fish species that commonly inhabit streams can demonstrate lower densities in reservoirs, with their abundances further decreasing downstream in inundated reaches (e.g. Gido *et al.*, 2002; Matthews, Gido & Gelwick, 2004), suggesting that reservoirs may be poor migration corridors for even generalist, small-bodied species.

The small-bodied habitat generalist *Cyprinella lutrensis* (Baird & Girard) experienced near, if not complete, extirpation in six of seven direct tributaries of Lake Texoma (Oklahoma/Texas, U.S.A.), whereas populations in the un-fragmented riverine networks upstream of the reservoir remained intact (Matthews & Marsh-Mathews, 2007). However, subsequent sampling (2008–2009) in one 'extirpated' direct tributary population (Brier Creek) revealed *C. lutrensis* had reappeared and then disappeared in two separate reaches of the stream (Marsh-Matthews, Matthews & Franssen, 2011). The declines of direct tributary populations and the failure of *C. lutrensis* to become reestablished were particularly surprising given *C. lutrensis* is hardy (Matthews & Hill, 1977), widespread (Matthews, 1987) and can numerically dominate fish assemblages in its native range (Marsh-Matthews & Matthews, 2000). Matthews & Marsh-Mathews (2007) and Marsh-Matthews *et al.* (2011) suggested increased predator densities, local habitat changes and reduced migration rates through reservoir habitats as possible mechanisms contributing to the decline in and reestablishment failure of *C. lutrensis* populations in direct tributaries.

Here, I assessed the genetic structure of the habitat generalist *C. lutrensis* from intact riverine and reservoir-fragmented stream populations in the Lake Texoma basin. I tested the prediction that the reservoir is acting as a barrier to gene flow among populations, assessed whether reduction in population sizes in direct tributaries has lowered genetic diversity in reservoir-fragmented populations and examined the potential population-of-origin of recently collected individuals in one previously 'extirpated' direct tributary population.

Methods

Study system and sampling

Denison dam impounded the Red and Washita Rivers in 1944, and formed Lake Texoma on the border of Oklahoma and Texas, U.S.A. (Fig. 1). Lake Texoma is a large (36 000 ha) and shallow (maximum depth 24 m) reservoir (Matthews *et al.*, 2004). Because the impoundment was constructed near the confluence of the Red and Washita Rivers, the reservoir has two distinct arms: the Red River and Washita River arms (Fig. 1).

Twelve sites were sampled in or near Lake Texoma (Table 1; Fig. 1). Specimens (n = 28–30) at each site were

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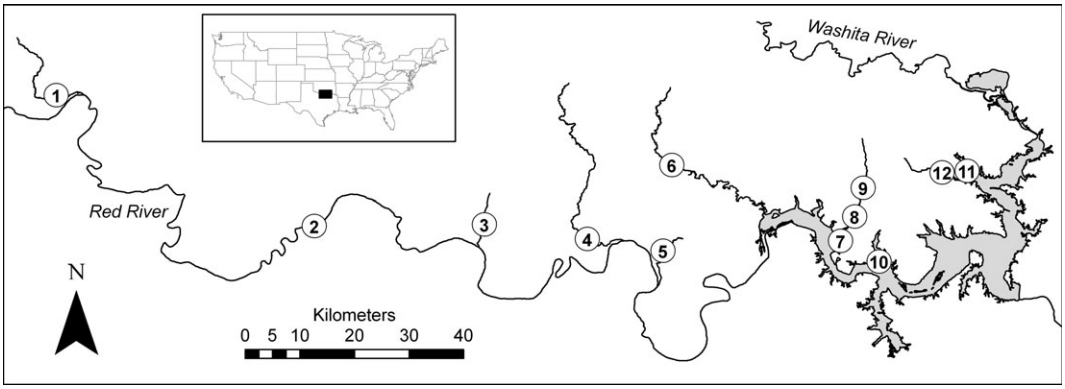

Fig. 1 Locations of sampling sites in the study area near Lake Texoma. Sites are coded by site numbers as stated in Table 1. Note the Red River flows west to east and the Washita River flows north-west to south-east and both debouch into Lake Texoma.

Table 1 Site location, site ID, date of collection, type of habitat, latitude and longitude of 12 sites where *Cyprinella lutrensis* individuals were collected to assess their genetic structure in and near Lake Texoma in 2008 and 2009

Site	ID	Date	Habitat	River Arm	Latitude	Longitude
Whiskey Creek	1	20 May 2008	River tributary	Red	34.1529	−98.1565
Red River	2	20 May 2008	River	Red	33.9377	−97.7320
Coffee Pot Creek	3	20 May 2008	River tributary	Red	33.9403	−97.4518
Walnut Bayou Creek	4	10 August 2009	River tributary	Red	33.9166	−97.2824
Bills Creek	5	10 August 2009	River tributary	Red	33.8970	−97.1577
Hickory Creek	6	19 May 2008	River tributary	Red	34.0377	−97.1434
Brier Creek Cove	7	23 June 2008	Reservoir	Red	33.9248	−96.8654
	7	30 July 2009				
Brier Creek Station 6	8	23 June 2009	Reservoir tributary	Red	33.9539	−96.8422
Brier Creek Station 5	9	4 June 2008	Reservoir tributary	Red	33.9990	−96.8286
Biostation Shore	10	12 June 2008	Reservoir	Red	33.8794	−96.8021
Glasses Creek Cove	11	12 August 2009	Reservoir	Washita	34.0281	−96.6611
Little Glasses Creek	12	2 August 2009	Reservoir tributary	Washita	34.0256	−96.6983

collected by seine and tissue was preserved in 95% ethanol in the field as whole individuals or as caudal fin clips and stored in 95% ethanol until DNA extraction. Six sites were in the un-fragmented Red River system upstream of Lake Texoma, three sites were in the reservoir proper (Red River arm $n = 2$, Washita arm $n = 1$) and three sites were in two direct tributaries of Lake Texoma ($n = 2$ in Brier Creek in the Red River arm, and $n = 1$ in Little Glasses Creek in the Washita River arm; Fig. 1). One site was sampled in both 2008 and 2009 (Brier Creek Cove; Table 1). In 2008 and 2009, sampling in Brier Creek at six different sites yielded sufficient *C. lutrensis* individuals for genetic analysis only at two sites, Brier Creek station 5 in 2008 and about 4 km downstream at Brier Creek station 6 in 2009. Collection of these individuals occurred following heavy rains and subsequent rises in the surface elevation of the reservoir, presumably inundating lower reaches of Brier Creek (Figure S1). No other direct tributaries of Lake Texoma (on the Oklahoma side) were included in genetic analyses because of the scarcity or absence of *C. lutrensis* in those habitats (see Matthews & Marsh-Mathews, 2007). Although Hickory Creek (site 6 in Fig. 1) appears to be a direct tributary of Lake Texoma, here it was considered a Red River tributary because this end of Lake Texoma has silted up over time and during elevated discharge the Red River flows past the Hickory Creek-Red River confluence.

Extraction of microsatellite DNA

DNA was extracted from dorsal muscle tissue from whole individuals or fin clips using a modified simple Chelex extraction (Walsh, Metzger & Higuchi, 1991): approximately 100 mg tissue and 300 μL of 10% Chelex solution were incubated at 99 °C for 12 min. Genetic variation was analysed at seven different microsatellite loci (Table S1). The forward primers were end-labelled with fluorescent dyes, and the polymerase chain reaction (PCR) was performed on a DNA Engine Dyad (MJ Research)

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thermocycler using two different multiplexed 12-μL reactions. The first reaction mixture contained 6.25 μL of Type-it Microsatellite Master Mix (Qiagen, Chatsworth, CA, U.S.A.), 2.5 μL of 50 μmol primers for three loci (Ca6, Nme 24B6.211 and Nme25C8.208), 0.5 μL template DNA and 4.5 μL dd$H_2$0. The second multiplexed PCR contained 6.25 μL of Type-it Microsatellite Master Mix, 1.25 μL of 50 μmol primers of four loci (Rhca20, Rhca24, Nme 18C2.178 and Nme 24B6.191), 0.5 μL template DNA and 3.25 μL dd$H_2$0. Thermocycler settings for both PCRs after denaturation at 95 °C for 5 min were 30 cycles of 95 °C for 30 s, 57 °C annealing temperature for 1.5 min followed by extension at 72 °C for 30 s. The final polymerisation step was extended to 30 min at 60 °C. PCR products were electrophoresed using a 3130xl Genetic Analyzer (Applied Biosystems, Foster City, CA, U.S.A.) with the GS600 size standard, and alleles were scored using PEAK SCANNER v1.0 software (Applied Biosystems).

Analyses

Variation in microsatellites. Preliminary scoring of alleles indicated one locus (Ca6) amplified products using PCR, but also amplified three alleles in some individuals and was therefore not included in further analyses. Microsatellite allele frequencies at each locus and population were tested for departures from Hardy–Weinberg equilibrium (Fisher's exact tests) and linkage disequilibrium between loci in GENEPOP ver. 4.0.10 (Raymond & Rousset, 1995). Probability tests were based on the Markov chain method with 10 000 dememorisations, 20 batches and 5000 iterations per batch.

Descriptive statistics and population divergence. General descriptive statistics [e.g. mean number of individuals amplified at each locus, observed and expected heterozygosities, and in-breeding coefficients (F_{IS})] were calculated in GENALEX ver. 6.4 (Peakall & Smouse, 2006) or FSTAT ver. 2.9.3.2 (Goudet, 1995). Genotypic frequencies of pairwise single and multilocus differences among populations (G-based tests) were tested using the permutation procedures with 10 000 dememorisations, 100 batches and 5000 iterations per batch in GENEPOP. Genetic divergence among populations was estimated using pairwise F_{ST} (= θ, Weir & Cockerham, 1984) with 10 000 permutations in ARLEQUIN ver. 3.5.1 (Excoffier & Lischer, 2010) and Reynolds coancestry coefficient (Reynolds, Weir & Cockerham, 1983) calculated with default settings in FSTAT. I used Analysis of Molecular Variance (AMOVA) with 1000 permutations in ARLEQUIN to assess the relative contribution of genetic variation attributable to within individuals, individuals within sites, and among sites. Where multiple tests were performed; all associated *P*-values were adjusted for multiple comparisons using a sequential Bonferroni correction (Holm, 1979; Rice, 1989).

I tested for patterns of isolation by distance and effects of the reservoir on genetic distance among populations using a partial Mantel test with 10 000 randomisations in FSTAT. The genetic distance matrix was pairwise F_{ST} between populations (with Brier Creek Cove 2008 and 2009 samples combined), and geographical distances between populations were based on stream segment and reservoir shoreline distances. The matrix used to assess the potential influence of reservoir habitat on genetic distances was coded as 1's and 0's where 1 denoted populations separated by reservoir habitat and 0 denoted populations separated by stream habitat.

To assess genetic structuring among populations, I used the program STRUCTURE ver 2.1 (Pritchard, Stephens & Donnelly, 2000) to identify genetically distinct clusters (*k*) following the method presented by Evanno, Regnaut & Goudet (2005). For each value of *k* (*k* = 1 through 13, i.e. the total number of collections), 10 iterations were run using the admixture model with a burn-in period of 100 000 iterations followed by 100 000 iterations in the collection phase. Each run was performed using an ancestry model incorporating admixture, a model of correlated allele frequencies and the prior population information as suggested by Pritchard *et al.* (2000).

Genetic diversity

Genetic diversity of populations was quantified using allelic richness (mean number of alleles per locus corrected for sample size) and Nei's gene diversity [the probability that, chosen at random, two copies of a gene (here, microsatellite loci) will be different alleles; Nei, 1987]. Only individuals that amplified all loci were included in these analyses. Nei's gene diversity was calculated in FSTAT using default settings. Mean allelic richness of each population was estimated using multiple random reductions (MRR; Leberg, 2002) in R with the package standArich (http://www.ccmar.ualg.pt/maree/software.php?soft=sarich). Multiple random reduction analysis is similar to rarefaction in that it estimates allelic richness accounting for sample size in individual samples (here each population). MRR resamples (with replacement) a subset of the individuals in each population, but sequentially samples (with several iterations) 1 through *n* individuals, where *n* is the total number of individuals in each population (Leberg, 2002). Mean allelic richness at

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each sample size interval was quantified from 100 iterations in each population. Mean numbers of alleles per locus was calculated from all individuals in each population whereas allelic richness of populations was compared at a sampling effort of 18 individuals (i.e. the smallest number of individuals that amplified all loci in one population).

Differences in allelic richness and gene diversity among loci and populations were tested using General Linear Models (GLM) with allelic richness or gene diversity as the dependent variable and population and locus as fixed factors using SPSS v. 18 (SPSS Inc., Chicago, IL, U.S.A.). The interaction term (population × locus) was not included because its addition would cause over-parameterisation of each model (i.e. each data point would be represented in the model once). Moreover, a significant interaction term would indicate allelic richness varied by loci among populations, which was not of particular interest here. *P*-values of pairwise comparisons in significant models were adjusted using the Bonferroni correction.

When testing the prediction that the three direct tributary populations would have reduced allelic richness because of isolating effects of the reservoir, the GLM suffered from low statistical power with using only five loci (see results below). To gain statistical power in testing this prediction, the observed allelic richness in the three direct tributary populations was compared to a null distribution of allelic richness generated by iterative subsampling of all genotyped individuals. This procedure estimates the probability that the observed allelic richness in the three direct tributaries were drawn from the same distribution of allelic richness predicted assuming no spatial structure in genetic diversity among populations. To generate the null distribution, allelic richness was quantified from a random selection of 18 individuals (with replacement) from all genotyped individuals with 1000 iterations using the standArich package in R. The probability that the observed mean allelic richness of the three direct tributary populations were a random subset of the null distribution was calculated using the Gaussian error function.

To test for recent bottlenecks in population sizes, the program BOTTLENECK (Cornuet & Luikart, 1997) was implemented using 1000 iterations with the Two Phase Model (TPM). The TPM is intermediate to the Stepwise Mutation Model (SMM) and Infinite Allele Model (IAM) and consists mostly of one-step mutations but 5–10% occur as multistep (Luikart *et al.*, 1998). Most microsatellite data sets investigated better adhere to the TPM than the IAM or SMM (Di Rienzo *et al.*, 1994). Probabilities of recent bottlenecks were assessed using the Wilcoxon sign-rank test.

Results

Variation in microsatellites

A total of 385 individuals was genotyped at six microsatellite loci. There was little evidence for departures from Hardy–Weinberg equilibrium at each locus in each population, except two loci deviated significantly from equilibrium in Bills Creek (Rhca24) and Walnut Bayou Creek (Rhca20), in both cases because of an excess of homozygotes. Two locus pairs showed significant linkage disequilibrium ($P < 0.05$) in Brier Creek 5 (Nme 24B6.191 and Nme 24B6.211) and Little Glasses Creek (Nme 24B6.211 and Rhc24). Therefore, the locus shared in both cases (Nme 24B6.211) was removed from further analyses. The remaining microsatellite loci ($n = 5$) were all polymorphic (mean = 19.80 ± 12.09 SD alleles per locus, range = 4–36) and demonstrated variation among populations (Table 2, Table S2). Most genetic variation was found within individuals (92%), followed by among individuals within populations (6.56%) and among populations (1.42%), suggesting a weak population structuring, overall.

Population divergence

When testing for divergence in allelic frequencies at each locus between all pairs of populations, G-based exact tests revealed significant divergence among populations for some of the five microsatellite loci independently (Table 3) and all loci combined (all $P < 0.05$). Calculation of F_{ST} values indicated some divergence among populations across loci (mean $F_{ST} = 0.015 \pm 0.018$ SD, range = −0.016–0.078; Table 3). Significant pairwise F_{ST} values were found primarily between population pairs that were separated by a great distance (e.g. Whiskey Creek, Red River and the two populations in the Washita arm of Lake Texoma) and five of the 11 pairwise comparisons with Brier Creek 5 (Table 3). Reynolds coancestry coefficients between population pairs also demonstrated a similar pattern (Table 3). The program STRUCTURE supported this divergence among some populations and suggested three genetic clusters coinciding with Brier Creek 5, the two populations in the Washita arm of the reservoir (Glasses Cove and Little Glasses Creek) and the remaining populations (Fig. 2).

Populations were significantly isolated by distance (Partial Mantel test, partial $r = 0.474$, $P < 0.001$; Fig. 3). Conversely, genetic distance did not correlate with the

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Table 2 Summary statistics of 12 populations of *Cyprinella lutrensis* collected in or near Lake Texoma between 2008 and 2009. Population names are the same as Table 1. Mean number of individuals genotyped per locus (N), mean number of alleles per locus (N_{all}), mean allelic richness per locus (A_r), mean gene diversity per locus (G_{div}), number of private alleles (P_{all}), mean expected (H_e) and observed heterozygosities (H_o) per locus, and the inbreeding coefficients averaged over all loci (F_{IS}) are presented

Population	ID	N	N_{all}	A_r	G_{div}	P_{all}	H_e	H_o	F_{IS}
Whiskey Creek	1	28.8	10.60	8.84	0.77	1	0.77	0.75	0.02
Red River	2	27.6	11.20	9.69	0.79	3	0.79	0.74	0.06
Coffee Pot Creek	3	29.6	12.20	10.24	0.80	2	0.80	0.74	0.07
Walnut Bayou Creek	4	29.4	11.00	9.27	0.77	1	0.77	0.66	0.14
Bills Creek	5	27.8	9.80	8.58	0.77	0	0.76	0.60	0.22
Hickory Creek	6	28.8	13.20	10.94	0.77	5	0.77	0.73	0.06
Brier Cove 2008	7	29.0	11.20	9.75	0.75	1	0.75	0.66	0.12
Brier Cove 2009	7	25.6	10.80	9.59	0.76	0	0.76	0.59	0.22
Brier Creek 6	8	25.4	11.20	10.60	0.81	2	0.81	0.70	0.14
Brier Creek 5	9	28.4	8.00	6.85	0.73	0	0.73	0.63	0.14
Biostation Shore	10	27.4	11.00	9.51	0.80	1	0.80	0.67	0.16
Glasses Creek Cove	11	27.4	9.40	7.94	0.71	0	0.71	0.64	0.10
Little Glasses Creek	12	29.6	11.40	9.80	0.78	1	0.78	0.71	0.09

Table 3 Summary data for spatial population genetic structure of *Cyprinella lutrensis* in the Lake Texoma basin (coded as in Table 1)

Population	1	2	3	4	5	6	7 (2008)	7 (2009)	8	9	10	11	12
1		0.006 0	0.012 0	0.011 0	0.009 0	0.039 1	0.025 0	0.015 0	0.008 0	0.057 4	0.006 0	0.081 1	0.070 1
2	0.006		0.001 0	0.001 0	0.005 0	0.021 0	0.010 0	0.017 0	0.000 0	0.041 0	0.000 0	0.043 1	0.037 1
3	0.012	0.001		0.003 0	0.000 0	0.006 0	0.009 0	0.003 0	0.000 0	0.035 2	0.000 0	0.016 0	0.017 0
4	0.011	0.001	0.003		0.000 0	0.009 0	0.000 0	0.005 0	0.000 0	0.035 1	0.000 0	0.028 0	0.032 0
5	0.009	0.005	−0.008	0.000		0.005 0	0.002 0	0.001 0	0.000 0	0.032 1	0.002 0	0.026 0	0.021 0
6	**0.038***	0.020	0.006	0.009	0.005		0.000 0	0.000 0	0.000 0	0.031 1	0.005 0	0.008 0	0.011 0
7 (2008)	0.025	0.010	0.009	−0.001	0.002	−0.003		0.001 0	0.002 0	0.040 1	0.009 0	0.021 0	0.027 0
7 (2009)	0.015	0.017	0.003	0.005	0.001	−0.001	0.001		0.004 0	0.035 0	0.006 0	0.030 0	0.029 0
8	0.008	0.000	−0.016	−0.002	−0.013	−0.006	0.002	0.004		0.027 2	0.000 0	0.019 0	0.002 0
9	**0.055***	**0.040***	**0.034***	0.034	0.032	0.030	**0.039***	0.034	0.026		0.023 1	0.050 1	0.034 1
10	0.006	−0.006	−0.008	−0.001	0.002	0.005	0.009	0.006	−0.016	0.023		0.035 0	0.020 1
11	**0.078***	**0.042***	0.016	0.027	0.026	0.008	0.020	0.029	0.019	0.048*	0.035*		0.000 0
12	**0.067***	**0.037***	0.016	**0.031***	0.021	0.011	0.027	0.028	0.002	0.033	0.020	−0.001	

The entries below the diagonal are pairwise F_{ST} values (bold text and asterisk indicate significant at sequential Bonferroni correction $P < 0.05$). Entries above the diagonal are Reynolds coancestry coefficient (above) and the number of loci with significant pairwise genotypic divergence over 5 loci (below).

separation of populations by reservoir habitats (partial $r = 0.123$, $P = 0.323$).

Genetic diversity

Allelic richness was variable among populations (Fig. 4), and the GLM demonstrated significant differences in allelic richness among loci ($F_{1,4} = 121.5$, $P < 0.001$) and populations ($F_{1,12} = 2.5$, $P = 0.014$). Bonferroni-corrected pairwise comparisons between populations revealed Brier Creek 5 had significantly lower allelic richness compared to only Brier Creek 6 and Hickory Creek. Gene diversity significantly differed among loci ($F_{1,4} = 135.2$, $P < 0.0001$) but not among populations ($F_{1,12} = 1.4$, $P = 0.194$). When

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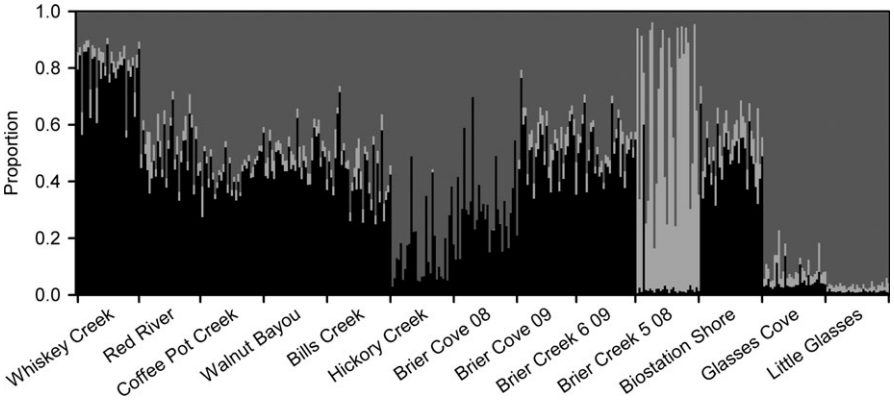

Fig. 2 STRUCTURE results showing three (k = 3) genetic clusters. Each vertical bar is an individual (grouped by population). Different colours in each bar demonstrate the proportion of times each individual was grouped in that cluster.

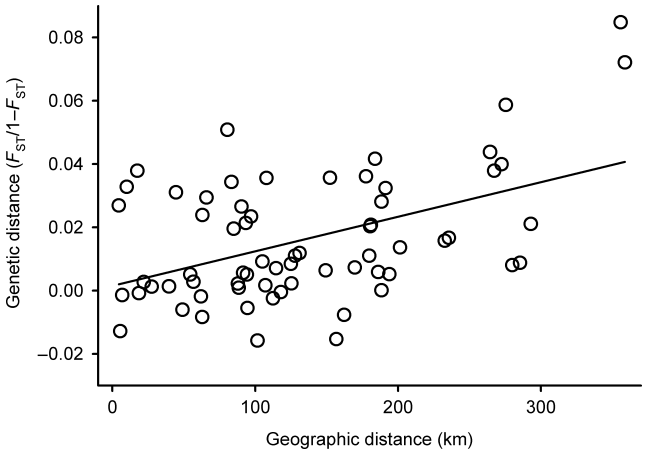

Fig. 3 Relationship between pairwise geographical distance (km) and genetic distance (F_{ST}/1–F_{ST}) among *Cyprinella lutrensis* in the Lake Texoma basin. The solid line denotes the best-fit linear regression model.

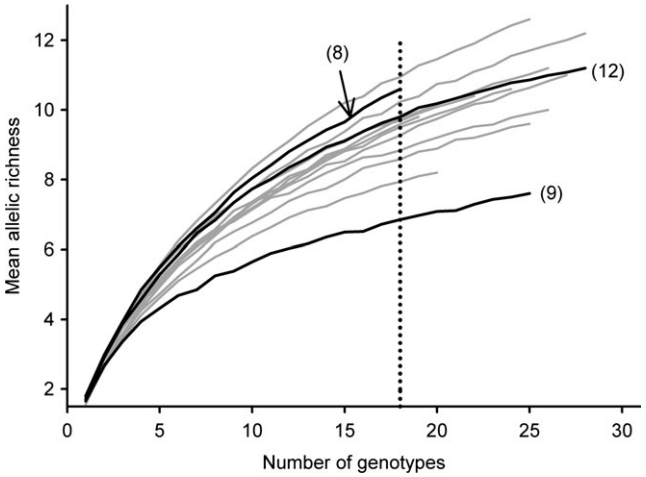

Fig. 4 Results of multiple random reductions for estimating mean allelic richness. Each ascending line is a population. The vertical dotted line at 18 genotypes indicates where allelic richness was compared among populations. The three direct tributary populations are noted: 8 = Brier Creek 6, 12 = Little Glasses Creek, 9 = Brier Creek 5. Brier Creek 5 had significantly lower allelic richness than predicted by the null distribution (see Fig. 5).

comparing observed allelic richness of direct tributary populations to the null distribution of expected allelic richness, Brier Creek 5 had significantly lower allelic richness than would be predicted by chance ($P < 0.001$) but not Brier Creek 6 ($P = 0.35$) or Little Glasses Creek ($P = 0.81$; Fig. 5).

There was no evidence for recent bottlenecks in population size among all populations (all $P > 0.05$) based on the results from BOTTLENECK.

Discussion

Genetic structuring of populations was primarily a function of isolation by distance, and most divergence was found between the two arms of the reservoir. One population in one direct tributary (Brier Creek 5) also showed increased genetic divergence that was probably not explained by geographical distances among populations. Genetic diversity (mean allelic richness) was also substantially lower in this population (Brier Creek 5) compared to all other populations.

Although many studies have assessed the effects of physical barriers (e.g. dams, waterfalls, weirs) on the genetic structure of fish populations (National Resource Council 1996), few have investigated the potential for reservoir habitats to act as barriers to gene flow (Skalksi *et al.*, 2008). Here, *C. lutrensis* (a habitat generalist) occurred in reservoir habitats, albeit in lower densities compared to nearby streams (Gido *et al.*, 2002; Matthews *et al.*, 2004). However, significant divergence among populations primarily occurred between the two arms of

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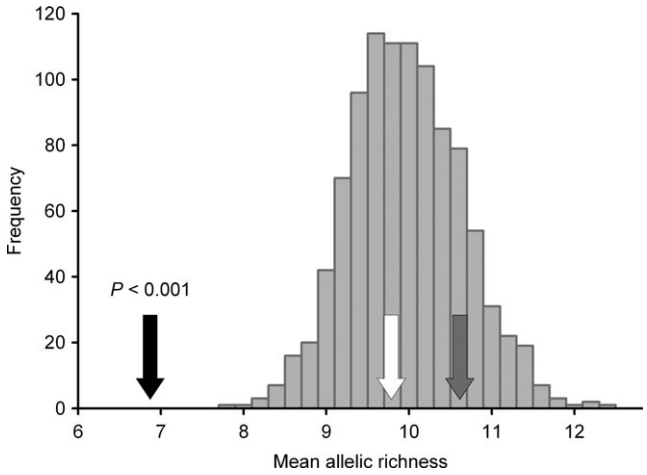

Fig. 5 The null distribution of mean allelic richness generated from all populations with a sample size of 18 individuals. Arrows indicate observed allelic richness in Brier Creek 5 (black arrow), Little Glasses Creek (white arrow), Brier Creek 6 (dark grey arrow). *P*-value indicates Brier Creek 5 had allelic richness significantly lower than predicted by the null distribution.

the reservoir and in only one (and at only one site: Brier Creek 5) of the two direct tributaries of Lake Texoma.

The significant isolation-by-distance correlation suggests a spatial structuring of *C. lutrensis* populations because of geographical distances among populations. However, the two populations from the Washita arm of Lake Texoma (i.e. Glasses Cove and the direct tributary, Little Glasses Creek) probably drove much of this relationship. Removal of these two populations resulted in a non-significant isolation-by-distance relationship among the remaining populations (Mantel test, $r = 0.27$, $P = 0.118$). The clustering of the two Washita arm populations based on the program STRUCTURE also suggests there is probably less gene flow between the two arms of the reservoir compared to gene flow within the Red River arm. This is not a surprising result given that the upstream reaches of reservoirs tend to resemble large rivers (Wetzel, 1990), and densities of small-bodied stream fish species can decrease further downstream in reservoirs (Gido *et al.*, 2002). Thus, low numbers of potential migrants in downstream reaches of the reservoir, coupled with reservoir habitats being potentially poor corridors for small-bodied fish migration (Schlosser *et al.*, 2000), may explain the genetic disparity between the two arms of the reservoir.

There was little evidence that gene flow was restricted between the direct tributary population and the reservoir in the Washita arm of Lake Texoma, as both populations were genetically similar. However, populations in the direct tributary (Brier Creek) in the Red River arm of the reservoir revealed genetic divergence compared to the nearest reservoir populations, but only one of the two sites within Brier Creek demonstrated this divergence. The furthest upstream site in Brier Creek (Brier Creek 5) was less similar compared to all other populations, including fish collected 4.5 km downstream at Brier Creek 6 and 10.0 km at Brier Creek Cove, suggesting some genetic isolation. This genetic disparity is unlikely to be a function of isolation by distance because of the close proximity of other populations, but the low *C. lutrensis* densities in Brier Creek (Matthews & Marsh-Mathews, 2007) coupled with recent lake level fluctuations may explain this longitudinal genetic structure.

Fish collected from Brier Creek 5 in 2008 were potentially from a relict population that has not exchanged genes with the reservoir for some time, whereas fish from Brier Creek 6 in 2009 were more likely recent immigrants from the reservoir proper. The low allelic richness observed in Brier Creek 5 suggests this population has potentially experienced genetic drift or founder effects, and the non-significant differences in Nei's gene diversity among populations was potentially because of the low statistical power with using only 5 loci. Conversely, Brier Creek 6 had the second highest observed allelic richness of all populations, suggesting these individuals have not suffered from genetic isolation. Increases in reservoir pool elevation can inundate lower reaches of direct tributaries in Lake Texoma (Matthews & Marsh-Mathews, 2007), potentially removing barriers to migration during periods of high water (e.g. Franssen *et al.*, 2006). The sudden appearance of individuals at Brier Creek 6 following a flood in May and June of 2009, and their genetic similarity to the reservoir proper populations, suggests these individuals were recent emigrants from the reservoir.

Divergence of some *C. lutrensis* populations suggests reservoir habitats may have impeded gene flow among populations. Yet it is unclear if, or to what extent, contemporary variation in allele frequencies were influenced by historical gene flow dynamics. Declines in direct tributary population sizes (Matthews & Marsh-Mathews, 2007), and subsequent genetic drift or founder effects probably contributed to the divergence observed in Brier Creek 5. However, the observed disparities between the arms of the reservoir could conceivably be a product of pre-impoundment isolation. Indeed, a pattern similar to the one observed here would arise if gene flow had been historically limited by physical or ecological barriers between populations in the Washita and the Red Rivers. However, the idea of historical barriers to gene flow between the two rivers seems unlikely given the rivers' large sizes, drainage of similar geographical regions, and

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the generalist and hardy nature of *C. lutrensis*. Nonetheless, the absence of data on the genetic structure of *C. lutrensis* populations prior to impoundment precludes certainty in the lack of historical barriers, and thus, the data presented here should be viewed in that light.

Investigations of stream habitat specialists in reservoir-fragmented systems may prove to be better candidates to assess fragmentation by reservoirs; however, results presented here suggest reservoir habitats may fragment populations of even the most generalist and hardiest of species. Continued investigations of the effects of reservoirs on native biota at larger spatial and temporal scales can be expected to prove useful for conservation managers of reservoir-fragmented ecosystems.

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Supporting Information

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Figure S1. (A) Mean daily surface elevation (m) above sea level of Lake Texoma from 2007 to 2010. The inset graph in (A) is the mean daily surface elevation (black line) and ± standard deviation (gray vertical bars) of Lake Texoma from 1995 through 2006. (B) Number of *Cyprinella lutrensis* collected from two sites in Brier Creek, a direct tributary of Lake Texoma between 2007 and 2010. Filled circles and solid lines are Brier Creek 5, and open circles and dotted lines are Brier Creek 6. Samples where tissue was collected for microsatellite analysis are indicated with (*).

Table S1. Locus, repeat type, primer sequences, multiplex reaction group, GenBank accession number, and source of microsatellite locus primer.

Table S2. Allele frequencies at each locus and population (coded as in Table 1).

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